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13. SUMMARY OF ENVIRONMENTAL OUTCOMES

13.1 Introduction

13.1.1 **Section 2** outlines the details and benefits of the Project together with the consideration of alternatives during the preliminary design stage, in which has taken into account of public concern and findings of various engineering and environmental review. The technical assessments in Sections 3 to 11 have demonstrated that all the statutory requirements in EIA Study Brief (ESB-375/2025) and EIAO-TM have been compiled with.

13.1.2 This chapter summarises the key environmental outcomes arising from this EIA study, the approaches that have been adopted to either avoid or minimise various environmental impacts throughout the design process, and the associated environmental enhancements and mitigation measures.

13.2 Environmental Benefits of the Project

13.2.1 This Project will provide a link road to connect the development within the Loop and KTN NDA. The development of the Project could result in the following benefit.

13.2.2 Northern Metropolis (NM) Synergy – According to Northern Metropolis Action Agenda, the Hong Kong-Shenzhen Innovation and Technology Park (the Park) in the Loop is identified as a key area in innovation and technology zone, and KTN NDA is identified as a key area for providing housing supply with more than 80,000 housing units in NM. This Project will act as a key linkage linking up these two major development nodes.

13.2.3 Enhance Traffic Connectivity – Currently, the Western Connection Road is the only external transport link connecting the Loop and the existing road networks. The proposed ECR can share the traffic demand for the Western Connection Road, and can strengthen the connectivity of the Loop and the road networks within KTN NDA via Ma Tso Lung area.

13.2.4 By adopting the preferred alignment of the proposed ECR which incorporates depressed roads and underpasses, the project minimises disruption to critical habitats including wetlands, fishponds, and flight corridors for migratory birds, as well as protected species like the Eurasian Otter. The use of natural ventilation for underpasses eliminates the need for energy-intensive mechanical systems, reducing long-term carbon emissions and visual impact. Additionally, the viaduct sections avoided direct impact to Shun Yee San Tsuen. During the operation phase, the ECR alleviates congestion on the WCR, lowering vehicular emissions from idling traffic and shortening travel times and results in further reducing fuel consumption and air quality impact.

13.2.5 Upon completion of the proposed ECR, the fishponds to be impacted by the alignment would be subject to ecological enhancement. Wetland associated Avifauna species, non-avifauna species of conservation species as well as Eurasian Otters would benefit from good water quality with better water management compared to existing fishpond, and increased riparian / emergent vegetation to provide suitable foraging habitat for these species.

13.3 Incorporation of Environmentally Friendly Options

13.3.1 Avoidance of environmental impacts is one of the main considerations throughout evaluation and development of alignment options. In case the environmental impacts could not be completely avoided due to limitation of actual site situations, environmentally friendly alternatives or designs were taken for compliance with relevant EIA requirements, subject to actual site condition/engineering design. The following approaches to avoid environmental impacts have been suitably incorporated into the layout of the proposed Alignment:-

- Avoidance of viaduct across the Meander: Direct impacts on the Meander are minimised, preserving wetland habitat integrity and maintaining ecological connectivity along the Meander for birds and other fauna.

- Minimised disturbance to Eurasian Otters and waterbirds: Eurasian Otters and other waterbirds in the Hoo Hok Wai–LMC area are sensitive to human activities and noise. The ECR alignment has been refined to minimise encroachment into major flight paths and nearby wetlands during the operation phase.

- Natural ventilation for the underpass: Incorporating natural ventilation reduces the need for large mechanical ventilation shafts, thereby limiting air quality, noise, and visual impacts to the surrounding environment.

- Limited ventilation openings: Construction of only two strategically located ventilation openings further minimises fishpond loss and reduces habitat fragmentation impacts.

13.4 Environmental Designs Recommended

13.4.1 Other than initiatives to avoid environmental impacts as summarised in **Section 13.3**, further efforts were exercised. The need for environmental designs required to mitigate the associated impacts were also identified and will be revisited and further established during detailed design for implementation as appropriate. A summary of these efforts is given below:-

- Adoption of environmentally friendly construction methods;

- Adoption of wildlife-friendly design; and

- Ecological Enhancement of Ponds 36 – 38

Adoption of Environmentally Friendly Construction Method

13.4.2 Precast construction method would be adopted as far as possible, with bridge deck formed from precast concrete sections which will be manufactured off site and joined together on site, which minimise the extent and duration of construction activities required on site and hence the potential environmental impacts in the vicinity during construction. Other construction methods would be considered only when precast construction method is not feasible.

Adoption of Wildlife-friendly Design

13.4.3 Provision of wildlife underpasses would be adopted to minimise impact of the Project on non-flying mammal species' movement in the impacted area and reduce potential wildlife injuries or mortality from roadkills. Design of the underpasses shall facilitate

potential wildlife movement. In addition, solid parapet as a standard road design along at-grade road, viaduct and depressed road would also help addressing potential roadkill impact.

Ecological Enhancement of Ponds 36 - 38

13.4.4 Ponds 36 – 38 would be reinstated and enhanced as ecologically enhanced pond upon completion of Phase 1 of the ECR Alignment and reinstatement works. The enhanced ponds are anticipated to function as an importance ecological linkage within the Deep Bay Wetland System, maintaining habitat connectivity and supporting the movement of wetland dependent fauna across the corridor, for example avifauna, Eurasian Otters and other terrestrial mammals.

13.5 Key Environmental Problems Avoided and Environmental Sensitivity Areas Protected

13.5.1 **Sections 13.3** and **13.4** summarise the key approaches adopted in the current design to avoid, minimise and mitigate environmental impacts. Some of these approaches have contributed to avoid a number of environmental problems and to protect a number of environmentally sensitive areas. **Table 13.1** presents the key environmental problems that have been avoided and any sensitive areas protected by these approaches.

Table 13.1 Summary of Key Environmental Problems Avoided and Sensitive Areas Protected

Design Approach	Environmental Problems Avoided and Sensitive Areas Protected
Adoption of environmentally friendly construction methods	• Adoption of off-site precast structures / modular integrated construction method for viaduct section to minimise the impacts on air quality, noise, water quality and waste during construction.
Adoption of wildlife friendly design to avoid potential wildlife injuries/mortality	• Provision of wildlife underpasses, and parapet along at-grade road, depressed road and viaduct to minimise habitat fragmentation disturbance and reduce wildlife injuries/mortality (e.g. roadkill).
Ecological Enhancement of Ponds 36 - 38	• Ecologically enhanced habitats to maintain habitat connectivity and support movement of wetland dependent fauna across the corridor of the Deep Bay Wetland System

13.6 Environmental Protection Measures Recommended

13.6.1 Mitigation measures were recommended to further reduce the environmental impacts due to construction and operation of the Project. Key recommended mitigation measures / good site practices / enhancement measures and their associated benefits are summarised in **Table 13.2** below.

Table 13.2 Key Recommended Mitigation Measures / Good Site Practices / Enhancement Measures and their Associated Benefits

Environmental Aspect	Key Recommended Mitigation Measures / Good Site Practices / Enhancement Measures	Associated Benefits
Air Quality Impact	Construction Phase: • Implementation of control measures as required in the Air Pollution Control (Construction Dust) Regulation to minimise construction air quality impact. • Implementation of control measures to minimise odour impact. • The Air Pollution Control (Non-road Mobile Machinery) (NRMMs) (Emission) Regulation and the Air Pollution Control (Fuel Restriction) Regulations would be followed and NRMMs supplied with mains electricity instead of diesel-powered should be adopted as far as possible to minimise the potential emission from NRMMs. Operation Phase: • No mitigation measures would be required.	• Protect Air Sensitive Receivers (ASRs) by reducing air pollutants emissions. • Ensure compliance with the requirements in Section 1 of Annex 4 of the EIAO-TM at the ASRs
Noise Impact	Construction Phase: • Implementation of quality and quieter powered mechanical equipment (PME), temporary / movable noise barriers and enclosure to screen off noise from relatively static PMEs, quieter construction methods, temporary movable noise barriers, noise enclosures, silencers as well as good site practices during construction phase of the Project to miminise construction noise impact. • For village houses / temporary structure with residential use which are close to the construction sites, it is recommended to install site hoarding with higher surface density and height to provide extra noise attenuation to protect these NSRs. • Submission of Construction Noise Management Plan (CNMP) before tendering. Operation Phase: • Implementation of Low Noise Road Surfacing (LNRS) and at-receiver	• Protect Noise Sensitive Receivers (NSRs) by reducing construction, fixed noise source and road traffic noise impacts. • Ensure compliance with the respective road traffic noise requirements at neighbouring NSRs.

Environmental Aspect	Key Recommended Mitigation Measures / Good Site Practices / Enhancement Measures	Associated Benefits
	mitigation measures such as acoustic windows/balconies to alleviate road traffic noise impact on surrounding NSRs. • Submission of Fixed Noise Sources Management Plan (FNMP) before tendering.	
Water Quality Impact	<u>Construction Phase:</u> • Construction of underpass road across the LMC Meander would require installation of cofferdam to enclose the excavation works. The construction should be conducted by phases and only one cofferdam should be installed in each stage. Cofferdams should be designed by the Contractor which fulfil the requirements in DSD Technical Circular No. 1/2017 "Temporary Flow Diversions and Temporary Works Affecting Capacity in Stormwater System". • Water pumps should be used to collect any construction site runoff and ingress/seepage water within the cofferdam to the on-site wastewater treatment facilities for treatment to satisfactory levels before discharge. • Silt curtain should be deployed to completely enclose the cofferdam prior to conduct cut-and-cover construction works in the LMC Meander, and installation and removal of cofferdams. • Mitigation measures and good site practices in ProPECCPN 2/24 “Construction Site Drainage • Practices in ETWB TC (Works) No. 5/2005 “Protection of natural streams / rivers from adverse impacts arising from construction works” • Waste Disposal Regulation • Provision of interim treatment facilities, such as chemical toilets, for construction workforce <u>Operation Phase:</u> • Stormwater Management Practices and Stormwater Pollution Control Plan to reduce non-point source pollution.	• minimise potential water quality impacts from construction activities, including construction works in the LMC Meander, construction site runoff, wastewater generated from general construction activities construction works near watercourses, diversion of watercourse, removal of ponds and wet areas, accidental spillage of chemicals, sewage from construction workforce, and groundwater infiltration arising from formation of underpass/depressed road • minimise potential water quality impacts from non-point source surface run-off

Environmental Aspect	Key Recommended Mitigation Measures / Good Site Practices / Enhancement Measures	Associated Benefits
	• Adequate design in silt trap for the new road drainage which take into account the guidelines in ProPECC PN 1/23 "Drainage Plan subject to Comments by the EPD".	
Waste Management Implications	Construction Phase: • Implementation of waste management hierarchy, good site practices, waste reduction measures and proper storage, collection and transportation of waste. Operation Phase: • No mitigation measures to be provided as the Project would not cause adverse impacts.	• Minimise waste generation • Reuse the excavated sediment. • Ensure the C&D materials are disposed at the designated locations.
Land Contamination	Subject to the finding of Contamination Assessment Plan / further site appraisal and Contamination Assessment Report (CAR), if contamination is identified, the remediation measures for remediation works as stated in Remediation Action Plan(s) should be implemented to ensure soil / groundwater contamination, if any, would be identified and properly remediated prior to construction works.	• any soil/groundwater contamination would be identified and properly treated prior to the construction works.
Ecology	Construction Phase: • Phased construction in wetland sensitive area including Zone 1, Zone 2 and Zone 3 as illustrated in **60722948/A10/Figure 3.2b**. • Wetland compensation in OWCAs, and enhancement in remaining area of Pond 36-38. • Off-site woodland compensation • Pre-construction survey to confirm the boundary of Seasonally Active Night Roost • Pre-construction survey and transplantation / translocation, and nest control measures for species of conservation importance • Pre-construction surveys for otter holts or natal dens • Time control (before 0900h and after 1700h during the period of October to April of the following year) of noisy construction activities (with the use of	• To mitigate the unavoidable loss of wetland habitats arising from the construction and operation of the Project. • To mitigate the unavoidable loss of woodland habitats arising from the construction and operation of the Project. • Minimising any direct / indirect impacts on flora species of conservation importance within the Project Boundary. • Minimising any potential direct / indirect impacts on

Environmental Aspect	Key Recommended Mitigation Measures / Good Site Practices / Enhancement Measures	Associated Benefits
	PME) within 100m buffer from the Seasonal Active Night Roost. • Time control (before 0900h and after 1700h) of aboveground noisy construction activities (with the use of PME) within wetland sensitive area including Zone 1, Zone 2 and Zone 3 as illustrated in **60722948/A10/Figure 3.2b**. • During construction, no above-ground structures exceeding 15m above the existing ground level are allowed in Zone 1, Zone 2, and Zone 3 as illustrated in **60722948/A10/Figure 3.2b** during the period from October to March of the following year. The maximum existing ground level in these zones is +5.5mPD; therefore, no above-ground structure may exceed +20.5mPD (i.e., +5.5mPD + 15m). • Provision of a temporary aboveground wildlife crossing to facilitate movement of non-flying mammals in impacted wetland area • Good site practices with mitigation measures for noise, lighting, dust, water quality impacts Operation Phase: • Wetland compensation in OWCAs, and wetland enhancement in Area 4 (the reinstated Pond 36-38) • Off-site woodland compensation • Provision of wildlife underpasses with solid parapet along the ECR. These underpasses would allow movement of non-flying mammals, reducing the risk of road collisions.	fauna species of conservation importance within and adjacent to Project Boundary. • Minimising any potential direct / indirect impacts on breeding / nesting activities of avifauna species of conservation importance within and adjacent to the Project Boundary. • Minimising impact of habitat fragmentation non-flying terrestrial mammals arose from the Project on within the vicinity of the Project. • Minimising potential roadkill of terrestrial wildlife at the ECR Alignment.
Fisheries	Construction Phase: • Maintaining bund stability • Minimisation of potential water quality impacts • Control of construction site run-off • Control of construction-related activities • Implementation of good site practices	• Minimising impact of construction works on the stability of fishpond bund and potential seepage from nearby fishponds into excavation area.

Environmental Aspect	Key Recommended Mitigation Measures / Good Site Practices / Enhancement Measures	Associated Benefits
Cultural Heritage	Construction Phase: • The monitoring for HB303 should be incorporated with a set of Alert, Alarm and Action (3As) system following AMO's monitoring requirements for grade 2 historic building, as well as vibration-sensitive and dilapidated buildings against continuous vibration. • An Authorized Person (AP) should be employed to carry out the condition survey and structural assessment to inspect the physical condition and structural integrity of the building and provide advice on the subsequent mitigation measures based on the assessment result. Operation Phase: • No mitigation measure would be required.	• To ensure no disturbance and physical damages made to the heritage sites during the course of works • To inspect the physical condition and structural integrity of the building
Landscape and Visual	Construction Phase: • Trees without impact from the proposed works should be retained and any existing trees unavoidably affected by the works should be transplanted as far as possible in accordance with DEVB TC(W) No. 4/2020 - Tree Preservation. • Erection of decorative screen hoarding or hoarding compatible with the surrounding setting to alleviate the landscape impact to surrounding public viewers. • Construction facilities and activities on work sites and areas should be carefully managed and controlled, including height and disposition / arrangement to minimise any potential adverse landscape impacts. • Reinstatement of the temporarily affected landscaped area within assessment area in like-for-like basis to restore the existing natural environment as far as possible. • Coordinated implementation programme with concurrent projects to minimise impacts and where possible reduce the period of disturbance due to the proposed works.	• To minimise the landscape impact on surrounding setting. • To blend in the structure to the adjacent environment and visual context. • To provide screening to and soften the hard edges of the proposed structures and mitigate the potential visual impact to key public viewers. • To compensate felled trees. • To compensate loss of fishponds. • Provision greenery on man-made engineered slope to mitigate the visual impact to the adjacent viewers and compensate the vegetation loss due to

Environmental Aspect	Key Recommended Mitigation Measures / Good Site Practices / Enhancement Measures	Associated Benefits
	Operation Phase: • Aesthetically pleasing design as regard to the form, material and finishes should be incorporated into bridges, ancillary buildings and other associated engineering facilities so as to blend in the structures with the adjacent landscape and visual context. • Buffer screen planting, including shrubs, to provide screening for the proposed structures if space is available. • As far as practicable, compensatory tree planting requirement of 1:1 in terms of number in accordance with DEVB TC(W) No. 4/2020 for compensation of felled trees subject to the availability of space and agreement made with relevant departmental parties. • Temporary loss of fishponds along ECR by the road widening and improvement works will be largely reinstated. These ponds will be used for both functional or amenity purposes to enhance the existing landscape and visual context. • Landscape treatments on slope or retaining structure in accordance with GEO Publication No. 1/2011 – Technical Guidelines on Landscape Treatment for Slopes shall be adopted subject to the availability of space and agreement made with relevant departmental parties	the slope cutting works.

13.7 Estimation of Population Protected

13.7.1 Population and environmental sensitive areas in the vicinity of the Project site have been protected through the avoidance and/or minimisation of environmental impacts from the construction and operation the Project. Population protected from air quality impacts include air sensitive receivers within 500m from the Project boundary. Population protected from noise impacts include noise sensitive receivers within 300m from the Project boundary including residential buildings, village houses, temporary structures. Population protected from water quality impacts include water sensitive receivers within 500m from the Project boundary.

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